

# Scale SEAL Leaderboards: Frontier Model Performance Across 20+ Bench

Comparative view of 100+ models across Agentic, Frontier, Safety, and Multimodal benchmarks

Source: Scale SEAL Leaderboards ([scale.com/leaderboard](https://scale.com/leaderboard))

# 20+ Benchmarks Spanning Four Capability Domains Evaluate 100+ Models

Scale SEAL spans Agentic, Frontier, Safety, and Multimodal benchmarks (private + public datasets; confidence intervals reported).

## **Agentic (coding + tool use)**

- SWE Atlas — Test Writing; Codebase QnA
- SWE-Bench Pro — Public; Private
- MCP Atlas — Tool Use

Focus: end-to-end software engineering and real tool invocation.

## **Frontier (reasoning + knowledge)**

- Humanity's Last Exam (HLE) + HLE (Text Only)
- EnigmaEval
- MultiChallenge; MultiNRC; SciPredict

Focus: hard multi-step reasoning; many scores far below saturation.

## **Safety (alignment + honesty)**

- MASK (Honesty)
- Fortress (National Security)
- PropensityBench (lower = better)

Focus: honesty/behavioral tendencies; performance varies sharply by task.

## **Multimodal (audio + vision-language)**

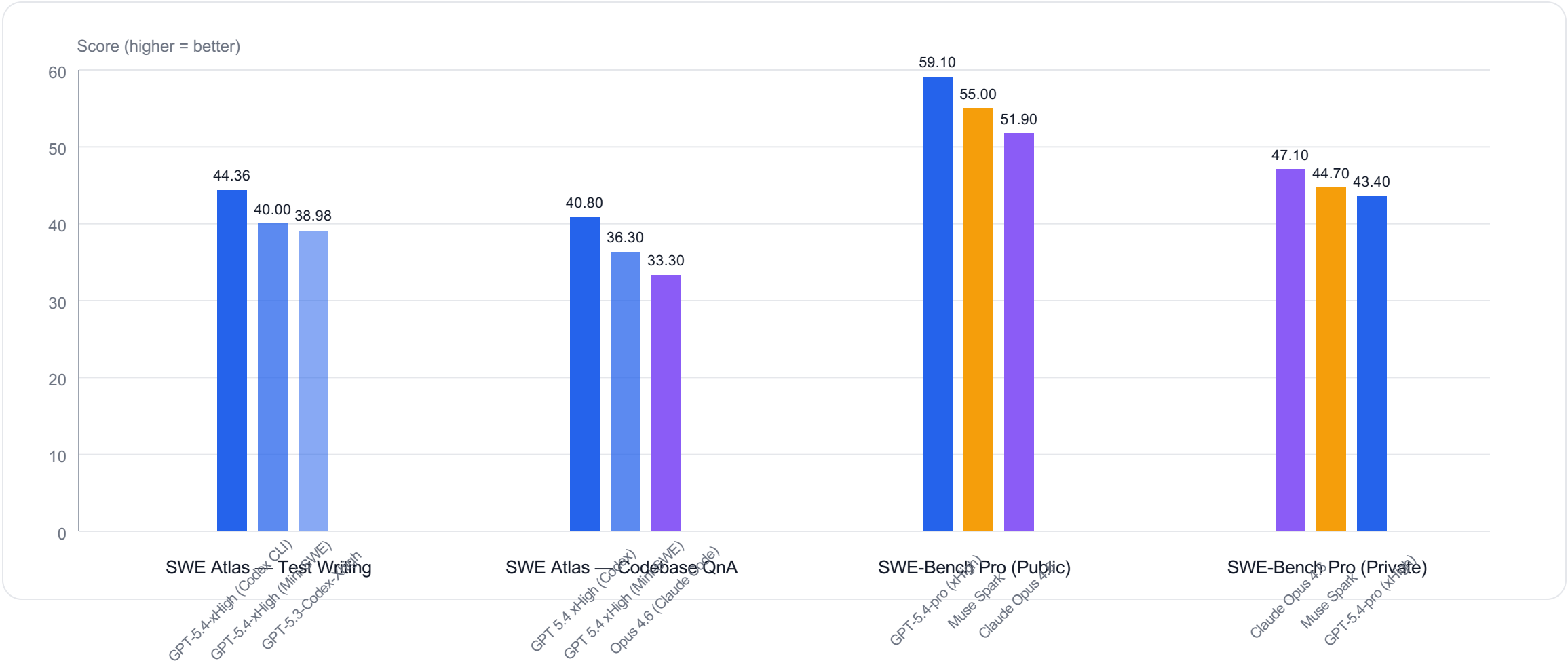
- AudioMultiChallenge (Text Output; Audio Output)
- VISTA (Vision-Language)
- VisualToolBench

Focus: robust audio + VLM capability; leaderboards dominated by Gemini variants.

# GPT-5.4 Dominates Agentic Coding: Top Scores on SWE Atlas and SWE-Bench

KEY INSIGHT: OpenAI leads SWE Atlas + SWE-Bench Pro (Public), while Claude Opus 4.6 leads SWE-Bench Pro (Private); Muse Spark consistently places top-3.

OpenAI Anthropic Google Muse Spark



# GPT-5.4-Pro Leads Humanity's Last Exam at 44.3% — Scores Remain Below

KEY INSIGHT: Frontier reasoning remains unsaturated — top HLE is ~44%, and EnigmaEval leaders are under 24%.

## Humanity's Last Exam (HLE)



## EnigmaEval



## MultiChallenge



# Gemini Models Sweep Audio and Vision-Language Benchmarks

KEY INSIGHT: Gemini variants lead AudioMultiChallenge (text + audio output) and VISTA, while GPT-5.4 stays competitive on VISTA and VisualToolBench.

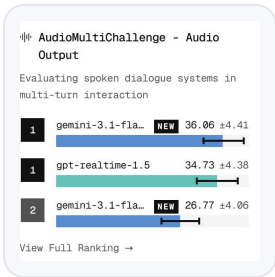
## AudioMultiChallenge

### Text Output



(All top-3 positions held by Gemini variants)

### Audio Output



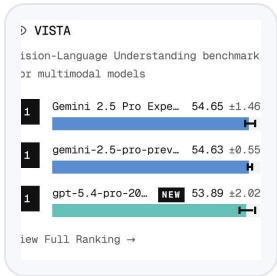
Reference: AudioMultiChallenge leaderboard (from brief)

## VISTA (Vision-Language)

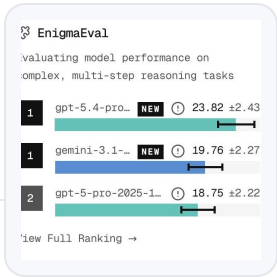


### VisualToolBench (note)

GPT-5.4 edges Gemini-3.1-pro: 29.17 vs 28.97



Reference: VISTA leaderboard (from brief)



Reference: EnigmaEval leaderboard (context for frontier slide)

# Claude Models Score 95–96% on MASK Honesty Benchmark; Fortress Score

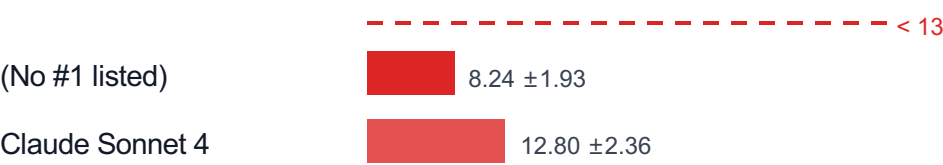
KEY INSIGHT: Alignment performance varies dramatically by task type — honesty is near ceiling, while Fortress remains in the low teens or single digits.

## MASK (Honesty) — near-ceiling scores



Interpretation: honesty benchmark is close to saturation for top Claude variants.

## Fortress (National Security) — low absolute scores



Interpretation: national-security benchmark remains difficult across models.

PropensityBench (lower = better) o3-2025-04-16: 10.50 ± 0.60

# OpenAI, Google, and Anthropic Each Claim #1 Across Different Domains

KEY INSIGHT: No single lab wins everywhere; leadership is benchmark-specific, with Muse Spark emerging as a top performer on select agentic + reasoning tasks.

| Legend                             |                               |
|------------------------------------|-------------------------------|
| OpenAI                             | Google                        |
| Anthropic                          | Muse Spark                    |
| Benchmark                          | #1 Model (Lab-coded)          |
| SWE Atlas – Test Writing           | GPT-5.4-xHigh (Codex CLI)     |
| SWE Atlas – Codebase QnA           | GPT 5.4 xHigh (Codex)         |
| MCP Atlas (Tool Use)               | Muse Spark                    |
| SWE-Bench Pro (Public)             | GPT-5.4-pro (xHigh)           |
| SWE-Bench Pro (Private)            | Claude Opus 4.6               |
| Humanity's Last Exam               | GPT-5.4-pro                   |
| HLE (Text Only)                    | GPT-5.4-pro                   |
| EnigmaEval                         | GPT-5.4-pro                   |
| MultiChallenge                     | Muse Spark                    |
| MultiNRC (Multilingual)            | GPT-5-pro                     |
| SciPredict                         | Gemini-3-pro-preview          |
| AudioMultiChallenge (Text Output)  | Gemini-3-pro-preview          |
| AudioMultiChallenge (Audio Output) | Gemini-3.1-flash-live-preview |
| VISTA (Vision-Language)            | Gemini 2.5 Pro Experimental   |
| VisualToolBench                    | GPT-5.4                       |
| MASK (Honesty)                     | Claude Opus 4.6               |
| PropensityBench (lower=better)     | o3-2025-04-16                 |
| Professional Reasoning – Legal     | Claude Opus 4.6               |
| Professional Reasoning – Finance   | Claude Opus 4.6               |
| Remote Labor Index (RLI)           | Claude Opus 4.6 (CoWo)        |

Note: Fortress omitted here because the brief does not specify a #1-ranked model.

# Key Takeaways: No Single Model Wins Everything — Specialization Is the N

- GPT-5.4 variants lead agentic coding and multiple frontier reasoning leaderboards — but not multimodal.
- Gemini dominates audio and vision-language benchmarks (AudioMultiChallenge + VISTA).
- Claude leads safety/honesty (MASK 96%+) and professional reasoning (Legal + Finance).
- Many benchmarks remain unsaturated: HLE ~44%, EnigmaEval ~24%, SciPredict ~25%.
- Muse Spark is a notable new entrant: #1 on MultiChallenge (75.52  $\pm$ 4.05) and MCP Atlas (78.30  $\pm$ 2.40), plus top-3 across 6+ benchmarks.

## Unsaturated benchmarks (top score)



Source: Scale SEAL Leaderboards (brief key findings + benchmark tables)

